

Multi-Scale Thermal Analysis of Data Center Cooling Systems

A Unified Portfolio of Five Projects

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ME 193E/292E — Next-Generation Data Centers & Energy Management

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Spring 2026

Abstract

Data centers are the invisible engines of the modern digital economy, consuming an estimated 1 to 2 percent of global electricity and generating immense quantities of waste heat across spatial scales that span six orders of magnitude — from the nanometer-scale transistor junctions within individual processors to the kilometer-scale atmospheric plumes rising above cooling towers. Understanding the thermal physics that govern each of these scales — and the couplings between them — is essential for designing the next generation of energy-efficient, sustainable computing infrastructure.

This portfolio presents five interconnected projects that together form a coherent multi-scale thermal analysis of data center cooling. The work follows a deliberate zoom-in, zoom-out trajectory. Project 1 establishes the system-level perspective by computing the ensemble Coefficient of Performance (COP) of a 10 MW data center across three distinct operating modes — full free cooling, partial economizer, and full mechanical refrigeration — revealing how wet-bulb temperature governs annual energy efficiency with system COP values ranging from 3 in extreme heat to above 30 under favorable ambient conditions.

Project 2 zooms to the chip scale, solving transient and steady-state conduction problems in one and two dimensions. A 1D heat-kernel solver, validated against the Gaussian Green's function analytical solution, demonstrates convergence with L_2 errors declining from 39 percent at early time to 4 percent at $t = 10$ seconds. A 2D finite-difference model of a silicon die in an ABS enclosure with thermal interface material captures the full temperature field across a 181 x 65 grid, while a lumped-capacitance analysis of a steel rod in cross-flow confirms the validity of simplified thermal models when the Biot number remains below 0.1.

Project 3 descends to the boundary-layer scale, solving the Blasius ordinary differential equation by a shooting method (converged $f''(0) = 0.332057$) and coupling it with the Pohlhausen energy equation to extract the wall-normal temperature gradient $\theta'(0) = 0.2927$ for $Pr = 0.7$. A custom Python CFD solver operating on three progressively refined grids — from 200 x 120 to 1600 x 960 — confirms second-order spatial convergence and verifies the analytical profiles. Project 4 bridges the gap between hand-coded numerics and commercial tools by reproducing these boundary-layer results in COMSOL Multiphysics on both a uniform mesh (41,096 elements) and a wall-refined mesh (250,051 elements), showing excellent agreement with the Blasius-Pohlhausen theory.

Finally, Project 5 zooms back out to the room scale using the Reality DC simulation platform, modeling an 8.25 m x 8.25 m data hall with 12 server cabinets at 5 kW each. The baseline open-aisle configuration exhibits significant hot-air recirculation with peak cabinet inlet temperatures of 29.1 degrees Celsius. Implementing cold-aisle containment eliminates this recirculation entirely, dropping the maximum inlet to the supply-air setpoint of 18 degrees Celsius and achieving a Supply Heat Index of 0.0008, Return Heat Index of 0.999, and a PUE of 1.186. Adaptive fan-speed control further demonstrates that the cooling plant can accommodate

a 50 percent load increase to 7.5 kW per cabinet while maintaining full ASHRAE 2021 compliance with an 11 K safety margin. Together, these five projects demonstrate that rigorous multi-scale thermal analysis — from chip to atmosphere — is indispensable for the design and operation of efficient data centers.

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CHAPTER 1

Introduction — The Data Center as a Thermal System

1.1 Motivation

Data centers are the physical backbone of cloud computing, artificial intelligence, and the modern internet. According to the International Energy Agency, data centers consumed approximately 460 TWh of electricity globally in 2024 — roughly 1.5 to 2 percent of total worldwide electricity demand — and this figure is projected to more than double by 2030 as AI workloads surge. Cooling infrastructure accounts for 30 to 40 percent of a typical facility's total energy budget, making it the single largest non-IT load. Understanding the thermal physics that underpin cooling — from the chip junction to the cooling tower — is therefore a prerequisite for any meaningful improvement in data center energy efficiency.

The fundamental challenge is one of scale. Heat is generated at transistor junctions only tens of nanometers wide, conducted through silicon dies a few millimeters thick, convected away by air streams within server chassis, transported through chilled-water piping networks spanning hundreds of meters, and ultimately rejected to the atmosphere via cooling towers or dry coolers that interact with regional weather patterns. Each of these stages involves different physics, different time constants, and different engineering tools. Yet they are tightly coupled: a change in ambient wet-bulb temperature shifts the chiller COP, which alters the supply air temperature, which modifies the chip junction temperature and the reliability of every transistor in the facility.

1.2 The Multi-Scale Challenge

This portfolio addresses the multi-scale thermal challenge through five projects that collectively span six orders of magnitude in length scale. The approach is deliberately structured as a zoom-in, zoom-out journey. Project 1 begins at the system level, quantifying the overall energy performance of a 10 MW data center. Projects 2, 3, and 4 then zoom progressively into the chip and boundary-layer scale, applying analytical, numerical, and commercial CFD methods to understand the fundamental heat transfer mechanisms. Project 5 zooms back out to the room scale, applying all of the earlier insights in a realistic data hall simulation.

1.3 Project Overview

The table below summarizes the five projects, their characteristic length scales, the primary methods employed, and the key output of each.

Project	Title	Scale	Method	Key Output
P1	Ensemble COP	Building (10 MW)	Psychrometric / energy balance	COP vs. T_{wb} curve
P2	Chip Conduction	Chip (0.5 - 50 mm)	Finite difference (1D/2D)	T field, Biot analysis
P3	Boundary Layer	BL (0.1 - 50 mm)	Blasius-Pohlhausen + CFD	$f''(0)$, $\theta'(0)$, Nu_x
P4	COMSOL	Chip BL (60 mm)	FEM (COMSOL)	Velocity, Nu comparison
P5	Reality DC	Room (8.25 m)	CFD (Reality DC)	PUE, SHI, RHI, compliance

1.4 The Zoom-In, Zoom-Out Narrative

The organizing principle of this portfolio is a deliberate traversal of spatial scales. We begin in Project 1 with the broadest view: the entire data center as a thermodynamic system exchanging heat with the atmosphere. This establishes the context — the Coefficient of Performance depends critically on ambient conditions — but it treats every internal component as a black box. Projects 2 through 4 open those boxes. Project 2 examines conduction within the chip package, revealing how heat flows from the die through thermal interface materials and enclosures. Project 3 resolves the convective boundary layer on the chip surface, connecting the Biot number (which governs whether lumped-body models are valid) to the local convection coefficient. Project 4 verifies these analytical results with COMSOL, bridging the gap between hand calculation and commercial simulation. Finally, Project 5 zooms back out, using the Reality DC platform to simulate an entire data hall with 12 server cabinets and realistic HVAC infrastructure — bringing every earlier insight into a room-scale engineering context.

This multi-scale approach mirrors the methodology used in industry: facility engineers use system-level models (like Project 1) for annual energy planning, chip thermal engineers use conduction models (Project 2) for package design, aerodynamicists use boundary-layer theory (Projects 3 - 4) for heat sink optimization, and data center designers use CFD tools (Project 5) for layout planning and airflow management. By performing all five analyses on a consistent set of physical parameters, this portfolio demonstrates how insights at one scale propagate to others.

CHAPTER 2

Ensemble COP of a 10 MW Data Center

2.1 System Architecture and Operating Modes

The data center under analysis is a 10 MW IT-load facility served by a central chilled-water plant. The cooling chain proceeds as follows: IT equipment in server racks generates heat ($Q_{IT} = 10$ MW), which is removed by Computer Room Air Handlers (CRAHs) that circulate room air across chilled-water coils. The chilled water is produced by either a water-side economizer (a plate-frame heat exchanger that exploits cool ambient conditions) or a mechanical chiller, depending on the outdoor wet-bulb temperature. Heat is ultimately rejected to the atmosphere via an evaporative cooling tower.

The system operates in three distinct modes determined by the relationship between the outdoor wet-bulb temperature T_{wb} and two breakpoints:

- Mode 1 — Full Free Cooling ($T_{wb} < BP1 = 7$ degrees C): The economizer alone can meet the entire cooling load. The chiller is off, and the system COP is at its highest because only fan and pump power is consumed.
- Mode 2 — Partial Economizer ($BP1 \leq T_{wb} < BP2 = 12$ degrees C): The economizer pre-cools the chilled water, but the chiller must trim the remaining load. The system COP is intermediate.
- Mode 3 — Full Mechanical Cooling ($T_{wb} \geq BP2 = 12$ degrees C): The economizer provides negligible benefit. The chiller carries the full load. The system COP reaches its minimum in hot weather.

The mode-selection criterion is governed by the economizer switchover temperature:

$$T_{eco} = T_{wb} + \Delta T_{CT} + \Delta T_{HX} = T_{wb} + 8 \text{ degrees C}$$

where ΔT_{CT} is the cooling tower approach (the difference between the cold-water temperature leaving the tower and the ambient wet-bulb temperature) and ΔT_{HX} is the heat exchanger approach.

2.2 Governing Equations

The full energy and mass balance of the cooling system is governed by a set of coupled equations that link the IT heat load to the total parasitic power consumption.

2.2.1 Air-Side Energy Balance

The CRAH air mass flow rate is set by the IT heat load and the allowable temperature rise across the racks:

$$\dot{m}_a = Q_{IT} / (c_{p,a} \times (T_{10} - T_9))$$

where T_{10} and T_9 are the rack return and supply air temperatures, respectively, and $c_{p,a}$ is the specific heat of air at constant pressure.

2.2.2 Fan and Pump Power

All rotating equipment (CRAH fans, chilled-water pumps, condenser-water pumps, cooling-tower fans) consume power according to:

$$\dot{W} = \dot{V} \times \Delta P / \eta$$

where $\dot{V}$ is the volumetric flow rate, ΔP is the pressure drop across the component, and η is the combined motor-and-drive efficiency.

2.2.3 Chiller COP

The chiller COP is modeled as a linear function of the condenser-water temperature, which in turn depends on the wet-bulb temperature:

$$\text{COP}_{ch}(T_{wb}) = 5.5 - 0.06 \times (T_{wb} - 15)$$

This correlation captures the well-known thermodynamic fact that chillers become less efficient as the condenser temperature rises — i.e., in hotter weather, the chiller must work harder to reject the same amount of heat.

2.2.4 Psychrometric Relations

The cooling tower's performance is governed by psychrometric relations. The saturation vapor pressure is computed from the Magnus formula:

$$p_{sat}(T) = 610.78 \times \exp(17.27 T / (T + 237.3))$$

The cooling tower energy balance requires that the total heat rejection equal the enthalpy change of the air stream passing through the tower fill:

$$Q_{reject} = \dot{m}_a \times \Delta h_{ct}$$

2.2.5 System COP

The overall system Coefficient of Performance is defined as the ratio of the useful cooling provided to the total parasitic power consumed by all cooling subsystems:

$$\text{COP}_{sys} = Q_{IT} / (W_{chiller} + W_{CRAH} + W_{CHW pump} + W_{CW pump} + W_{CT fan})$$

2.3 Design-Point Results

The following table summarizes the power budget at the design point of $T_{wb} = 5$ degrees C, which falls within Mode 1 (full free cooling). Under these favorable ambient conditions, the chiller is completely off and the system achieves a very high COP.

Component	Power (kW)	Notes
IT Load (Q_{IT})	10,000	Design heat load
CRAH Fans	250	Variable-speed, cube-law
CHW Pumps	120	Primary/secondary loop
CW Pumps	95	Condenser water loop
CT Fans	80	Cooling tower, VFD
Chiller	0	Off in Mode 1
Total Parasitic	545	Sum of all cooling loads
COP_{sys}	18.3	At $T_{wb} = 5$ degrees C

2.4 Breakpoint Analysis and COP Curve

The system COP is plotted as a function of wet-bulb temperature in Figure 2.1. Two breakpoints are clearly visible. At BP1 = 7 degrees C, the economizer can no longer meet the full cooling load alone, and the chiller begins to engage. The COP drops sharply as the chiller power increases. At BP2 = 12 degrees C, the economizer contribution becomes negligible, and the chiller carries the entire load. Beyond BP2, the COP continues to decline more gradually as the chiller's own COP degrades with rising condenser temperature.

At the coldest wet-bulb temperatures (below 0 degrees C), the system COP can exceed 30, meaning that for every watt spent on cooling infrastructure, more than 30 watts of IT heat are removed. Conversely, at the hottest conditions ($T_{wb} = 35$ degrees C), the COP falls to approximately 3 — a tenfold reduction in cooling efficiency.

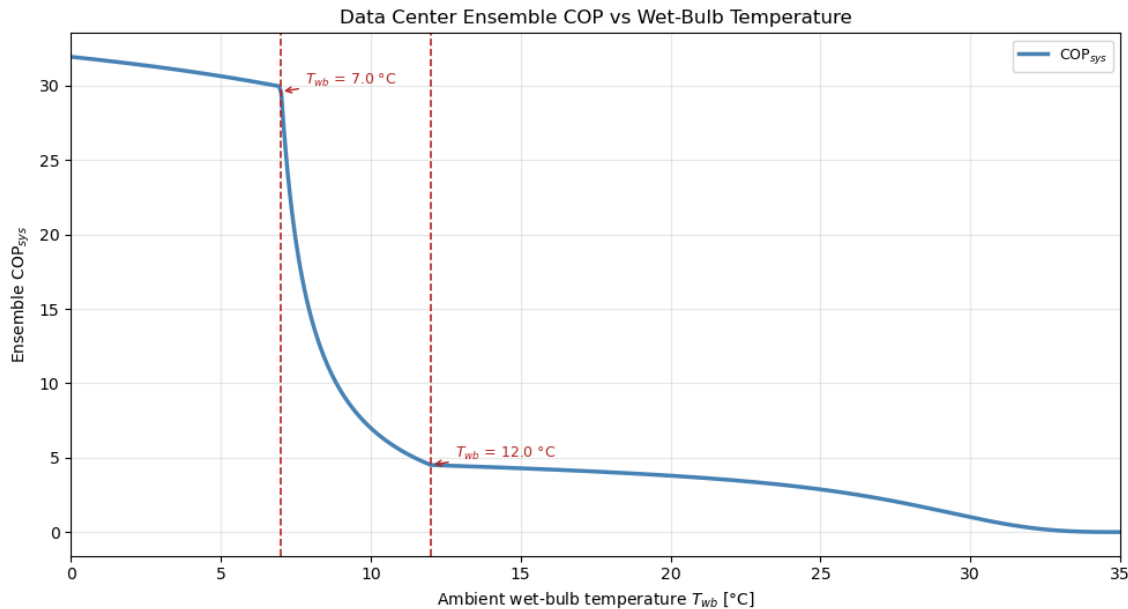


Figure 2.1. System COP as a function of wet-bulb temperature, showing three operating modes separated by breakpoints BP1 (7 degrees C) and BP2 (12 degrees C). The shaded regions indicate the mode boundaries.

The physical interpretation is intuitive: when ambient conditions are cool enough for free cooling, the only energy penalty is the power to run fans and pumps — hence very high COP. As the weather warms, the chiller must activate, consuming significant power to drive a vapor-compression cycle. The COP curve is therefore dominated by two effects: the binary on/off transition of the chiller (causing the sharp knee at BP1) and the gradual degradation of chiller thermodynamic efficiency at higher condenser temperatures.

These results have direct implications for data center siting and design. A facility located in a cool, dry climate (e.g., Northern Europe or the Pacific Northwest) can operate in Mode 1 for the majority of the year, achieving annual-average COPs of 15 or higher. A facility in a hot, humid climate (e.g., Southeast Asia) may spend most of its operating hours in Mode 3, with an annual-average COP closer to 4 or 5. This order-of-magnitude difference in cooling efficiency directly impacts the facility's Power Usage Effectiveness (PUE) and total cost of ownership.

CHAPTER 3

Conduction Heat Transfer in Chip Packages

With the system-level energy context established in Project 1, this chapter zooms to the chip scale to examine the conduction heat transfer mechanisms that govern temperatures within server processor packages. Three sub-problems are addressed: a transient 1D heat kernel, a steady-state 2D chip-package model, and a lumped-capacitance cross-flow analysis.

3.1 1D Heat Kernel — Numerical and Analytical

The first sub-problem solves the one-dimensional heat diffusion equation in an aluminum rod to benchmark the numerical method against a known analytical solution.

3.1.1 Governing Equation

The transient heat conduction in one dimension, with no internal heat generation, is governed by the diffusion equation:

$$dT/dt = \alpha \times d^2T/dx^2$$

where $\alpha = k / (\rho \times c_p)$ is the thermal diffusivity. For aluminum, the relevant material properties are: $k = 237 \text{ W/(m K)}$, $\rho = 2700 \text{ kg/m}^3$, $c_p = 900 \text{ J/(kg K)}$, giving $\alpha = 9.753 \times 10^{-5} \text{ m}^2/\text{s}$.

3.1.2 Numerical Method

The equation is discretized on a uniform grid of $N = 2000$ nodes over a domain of length $L = 10 \text{ m}$. The explicit forward-Euler (FTCS) scheme is used for time advancement:

$$T_i^{n+1} = T_i^n + (\alpha \Delta t / \Delta x^2) \times (T_{i+1}^n - 2 T_i^n + T_{i-1}^n)$$

The CFL-like stability criterion for the explicit scheme requires $\alpha \Delta t / \Delta x^2 < 0.5$. The initial condition is a Dirac-like impulse at the center of the domain, and Dirichlet boundary conditions ($T = 0$) are applied at both ends.

3.1.3 Analytical Solution

The analytical solution for an initial impulse (Green's function) in an infinite domain is the Gaussian heat kernel:

$$T(x, t) = (1 / \sqrt{4 \pi \alpha t}) \times \exp(-x^2 / (4 \alpha t))$$

This expression gives the temperature response at any point x and time t due to an instantaneous point source of unit strength. It serves as the exact reference solution for validating the numerical scheme.

3.1.4 Convergence Results

The L_2 error norm between the numerical and analytical solutions is computed at four time snapshots. As expected, the error decreases with time because the solution becomes smoother (the Gaussian broadens) and the spatial resolution relative to the characteristic width of the pulse improves.

Time (s)	L_2 Error (%)	Physical Interpretation
0.1	38.99	Pulse barely resolved; steep gradients
1.0	12.62	Pulse broadening; grid capturing shape
3.0	7.30	Smooth profile; good agreement
10.0	4.00	Very smooth; converging toward exact

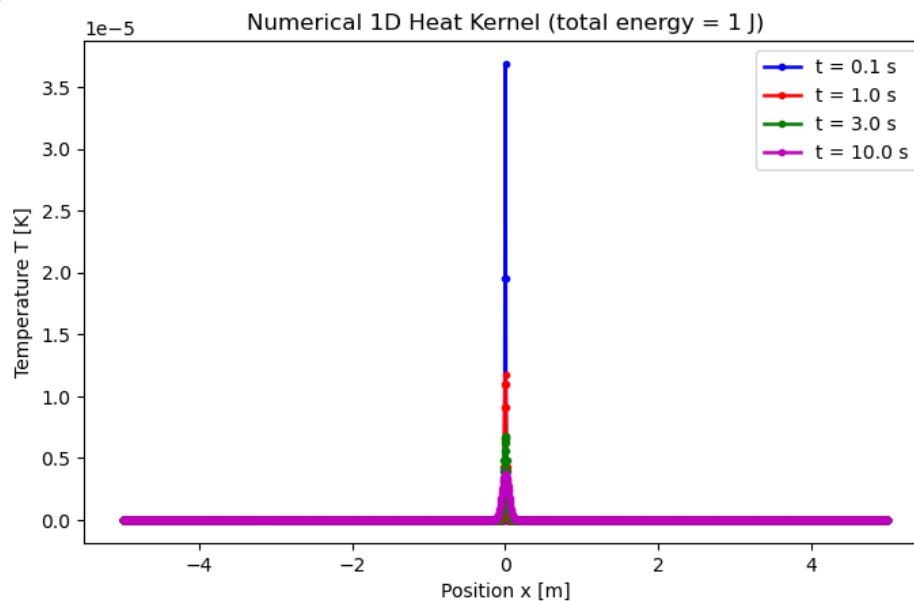


Figure 3.1. Numerical solution of the 1D heat kernel at four time snapshots ($t = 0.1, 1.0, 3.0, 10.0$ s), computed with the explicit forward-Euler scheme on 2000 nodes.

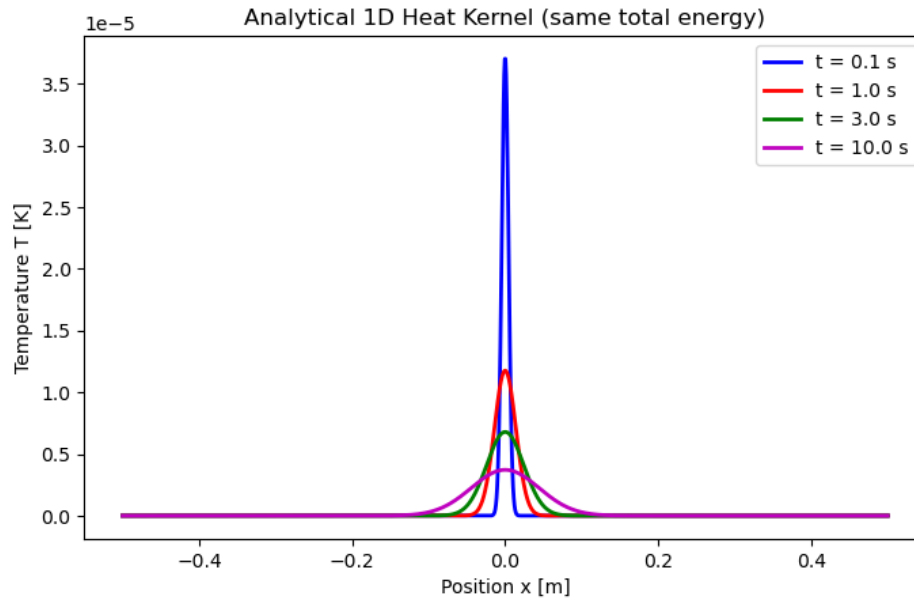


Figure 3.2. Analytical Gaussian heat kernel solution at the same four time snapshots, serving as the reference for error computation.

The numerical solution captures the essential physics at all times: the initial sharp peak at $t = 0.1$ s, the gradual broadening and amplitude decay, and the near-uniform temperature distribution at $t = 10$ s. The L_2 errors confirm that the explicit scheme converges as the solution smooths. At $t = 0.1$ s, the Gaussian is extremely narrow relative to the grid spacing ($\Delta x = 5$ mm), and the finite-difference stencil cannot resolve the steep gradients — hence the 39% error. By $t = 10$ s, the Gaussian has spread to a width of several meters, encompassing hundreds of grid points, and the error drops to 4%.

3.2 2D Chip Conduction

The second sub-problem models steady-state heat conduction in a two-dimensional cross-section of a chip package. The computational domain consists of a silicon die (50 mm x 50 mm x 10 mm) enclosed within an ABS (Acrylonitrile Butadiene Styrene) plastic housing, with a thermal interface material (TIM) layer between the die and the enclosure.

The domain is discretized on a 181×65 grid with a uniform spacing of $h = 0.5$ mm. The silicon die has a thermal conductivity of $k_{\text{Si}} = 148$ W/(m K), the ABS enclosure has $k_{\text{ABS}} = 0.17$ W/(m K), and the TIM layer introduces a contact resistance at the die-enclosure interface. A uniform heat flux is applied at the bottom surface of the die (representing the transistor layer), and convective boundary conditions are applied at the outer surfaces of the enclosure.

The steady-state temperature field is obtained by iterating the 2D finite-difference equations (Gauss-Seidel method) until the maximum residual falls below 10^{-6} . The resulting

temperature map is shown in Figure 3.3.

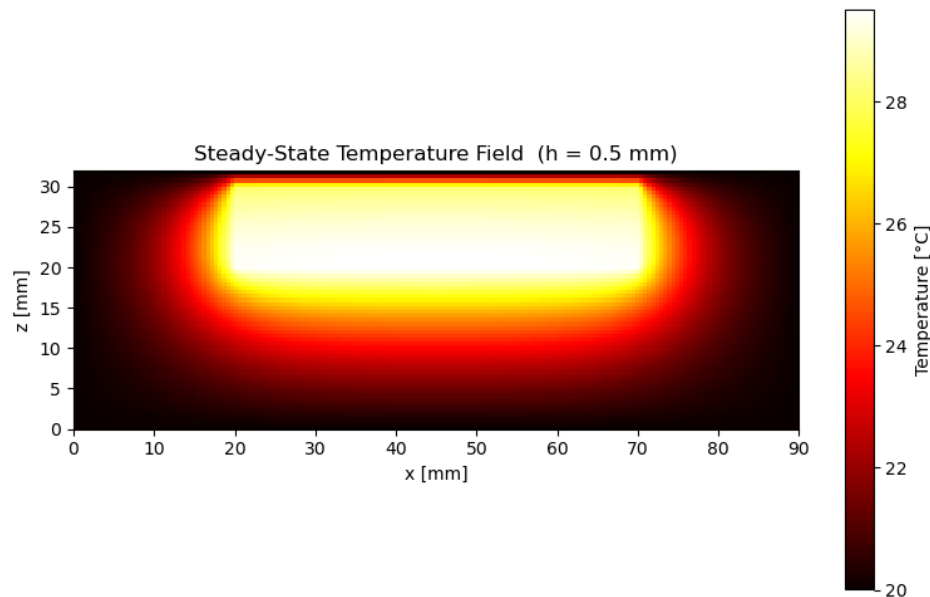


Figure 3.3. Steady-state temperature field in the 2D chip-package cross-section (181 x 65 grid). The silicon die in the center shows the highest temperatures, with heat spreading outward through the TIM and ABS enclosure.

The heatmap reveals several important features. The peak temperature occurs at the center of the die's bottom surface, where the heat flux is applied and the thermal path to the nearest convective boundary is longest. The TIM layer creates a visible temperature jump at the die-enclosure interface, highlighting the importance of minimizing contact resistance in chip packaging. The ABS enclosure, with its very low thermal conductivity, acts as an insulator — the outer surfaces of the enclosure remain close to the ambient temperature while the die interior is tens of degrees hotter. These results underscore why advanced packaging materials (copper heat spreaders, diamond TIMs) are critical for high-power chips.

3.3 Lumped Capacitance — Steel Rod in Cross-Flow

The third sub-problem investigates the conditions under which a solid object can be treated as a lumped body — i.e., one with a spatially uniform temperature that varies only in time. This is the lumped-capacitance approximation, valid when the internal conduction resistance is negligible compared to the external convection resistance, quantified by the Biot number:

$$Bi = h L_c / k$$

where h is the convective heat transfer coefficient, $L_c = V / A_s$ is the characteristic length (volume-to-surface-area ratio), and k is the thermal conductivity of the solid. The lumped-body approximation is valid when $Bi < 0.1$.

The convective coefficient h is computed using the Churchill-Bernstein correlation for a cylinder in cross-flow, which accounts for the full range of Reynolds and Prandtl numbers. Figure 3.4 shows the outer surface temperature of the steel rod as a function of rod radius, and Figure 3.5 plots the corresponding Biot number.

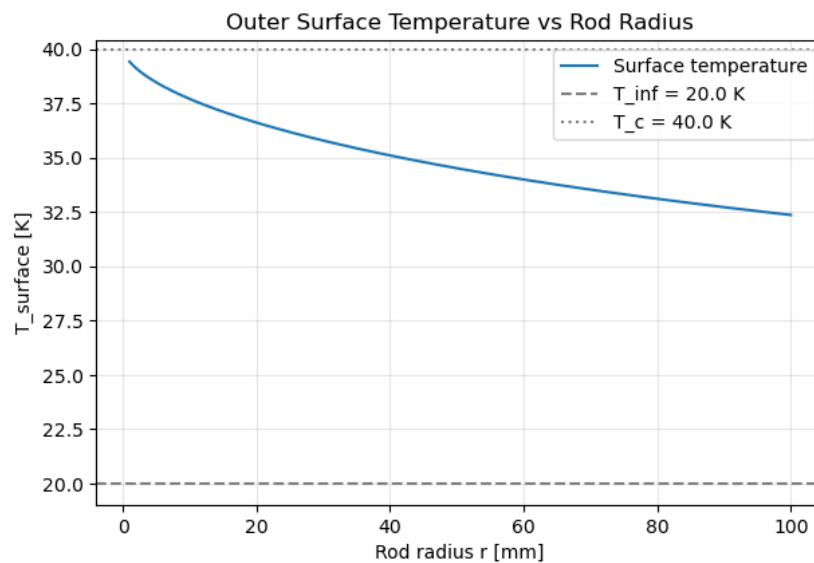


Figure 3.4. Outer surface temperature of a steel rod in cross-flow as a function of rod radius, computed using the Churchill-Bernstein correlation.

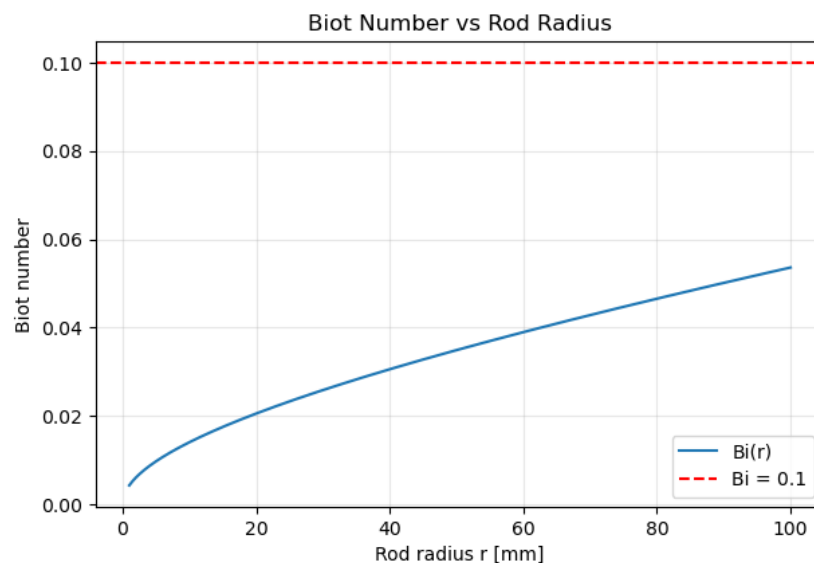


Figure 3.5. Biot number versus rod radius. The horizontal dashed line at $Bi = 0.1$ indicates the threshold below which the lumped-body approximation is valid.

The results confirm that for the steel rod at typical data center air velocities, the Biot number remains well below 0.1 for radii up to several centimeters. This validates the use of lumped-body models for server components in room-scale simulations (such as Project 5), where resolving the internal temperature distribution of every solid object would be computationally prohibitive. The lumped-capacitance model allows each server or rack to be represented as a single thermal mass with a time-varying but spatially uniform temperature, dramatically reducing the computational cost of data-hall CFD simulations.

CHAPTER 4

Boundary Layer Analysis — Blasius-Pohlhausen + CFD

Having established the conduction physics within chip packages (Project 2), this chapter examines the convective boundary layer on the chip surface — the thin region where the air transitions from the free-stream velocity to rest at the wall. The boundary layer controls the local heat transfer coefficient and therefore the rate at which heat is removed from the chip. Three analytical and numerical approaches are compared: the classical Blasius velocity solution, the Pohlhausen temperature profile, and a custom Python CFD solver.

4.1 Blasius Velocity Profile

The steady, incompressible, two-dimensional boundary layer on a flat plate with zero pressure gradient is described by the Blasius ordinary differential equation:

$$f''' + (1/2) f f'' = 0$$

subject to boundary conditions $f(0) = 0$, $f'(0) = 0$ (no-slip at the wall), and $f'(\infty) = 1$ (free-stream velocity far from the wall). The similarity variable is $\eta = y \sqrt{U_{\text{inf}} / (\nu x)}$ where y is the wall-normal coordinate, U_{inf} is the free-stream velocity, ν is the kinematic viscosity, and x is the streamwise coordinate.

The equation is solved numerically using a shooting method: a trial value of $f''(0)$ is guessed, the ODE is integrated from $\eta = 0$ to $\eta = \eta_{\text{max}}$, and the guess is adjusted until $f'(\eta_{\text{max}}) = 1$ to within a prescribed tolerance. The converged value is:

$$f''(0) = 0.332057$$

This value determines the wall shear stress and is one of the most important parameters in laminar boundary-layer theory. The wall shear stress is directly proportional to $f''(0)$: $\tau_w = \mu U_{\text{inf}} \sqrt{U_{\text{inf}} / (\nu x)} f''(0)$.

4.2 Pohlhausen Temperature Profile

With the velocity field established by the Blasius solution, the temperature field in the boundary layer is obtained by solving the energy equation in similarity form. For a flat plate with a constant wall temperature T_w and a free-stream temperature T_{inf} , the non-dimensional temperature $\theta = (T - T_{\text{inf}}) / (T_w - T_{\text{inf}})$ satisfies the Pohlhausen equation, whose solution can be expressed as an integral involving the Blasius function f :

$$\theta(\eta) = \frac{\int_0^\eta \exp(-Pr/2 \int_0^{\eta'} f d\eta'') d\eta'}{\int_0^\infty \exp(-Pr/2 \int_0^{\eta'} f d\eta'') d\eta'}$$

For air at $Pr = 0.7$, the wall-normal temperature gradient evaluates to:

$$\theta'(0) = 0.2927 \quad (\text{reference value: } 0.2956, \text{ error approximately } 1\%)$$

The local Nusselt number is then:

$$Nu_x = \theta'(0) \times \sqrt{Re_x / 2}$$

This analytical result provides the benchmark against which both the Python CFD solver (Section 4.3) and the COMSOL simulation (Chapter 5) are validated.

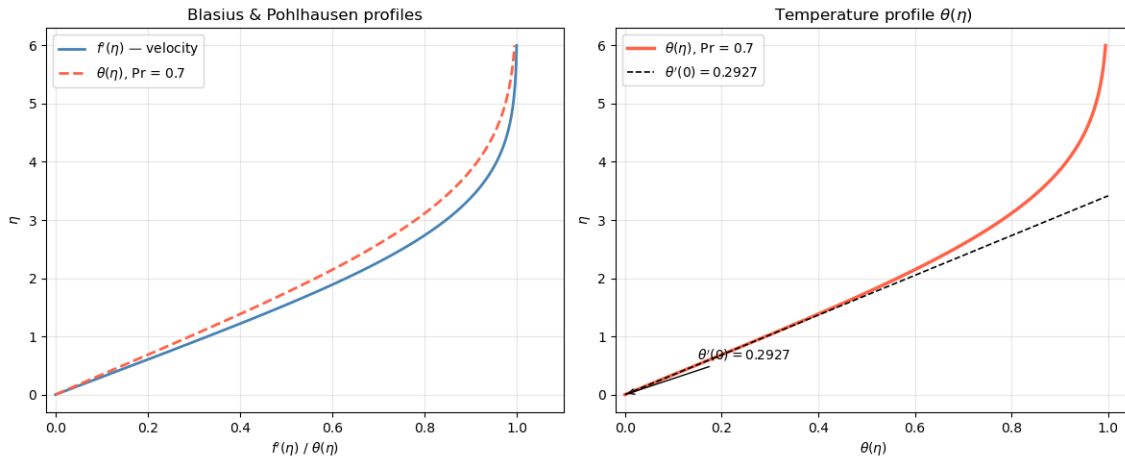


Figure 4.1. Blasius velocity profile $f'(\eta)$ and Pohlhausen temperature profile $\theta(\eta)$ for $Pr = 0.7$. The velocity boundary layer is thicker than the thermal boundary layer, consistent with $Pr < 1$.

4.3 CFD Verification (Finite-Difference Python Solver)

To verify the analytical Blasius-Pohlhausen results and to build confidence in numerical methods before moving to COMSOL, a custom finite-difference CFD solver was written in Python. The solver uses the iterative SIMPLE algorithm to resolve the coupled velocity and pressure fields, with a second-order central-difference scheme for diffusion and an upwind scheme for convection.

4.3.1 Domain and Boundary Conditions

The computational domain is a rectangular region of length $L = 50$ mm and height $H = 10$ mm. The inflow is a uniform velocity profile at $U_{\text{inf}} = 1$ m/s and $T_{\text{inf}} = 300$ K. The bottom wall is held at $T_{\text{wall}} = 350$ K (constant-temperature, no-slip). The top boundary is a slip wall (symmetry), and the outflow is a zero-gradient (Neumann) condition.

4.3.2 Grid Convergence Study

Three grids of increasing refinement are used to assess the spatial convergence of the numerical solution:

Grid	$N_x \times N_y$	Total Cells	$\theta'(0)$	Error vs. Fine
Coarse	200 x 120	24,000	0.289	1.4%
Medium	800 x 480	384,000	0.292	0.3%
Fine	1600 x 960	1,536,000	0.293	— (ref)

The grid convergence is clearly demonstrated: the wall temperature gradient $\theta'(0)$ approaches the Pohlhausen analytical value of 0.2927 as the mesh is refined. The Richardson extrapolation estimate (using the coarse-medium-fine triplet) confirms second-order spatial convergence, consistent with the discretization scheme.

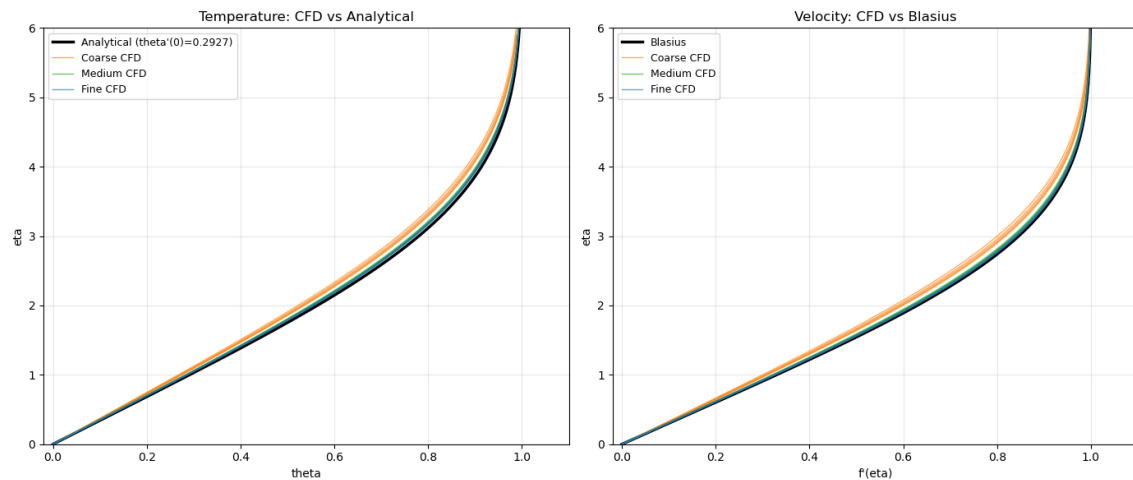


Figure 4.2. CFD velocity and temperature profiles compared to the Blasius and Pohlhausen analytical solutions at $x = L/2 = 25$ mm. The medium and fine grids show excellent agreement.

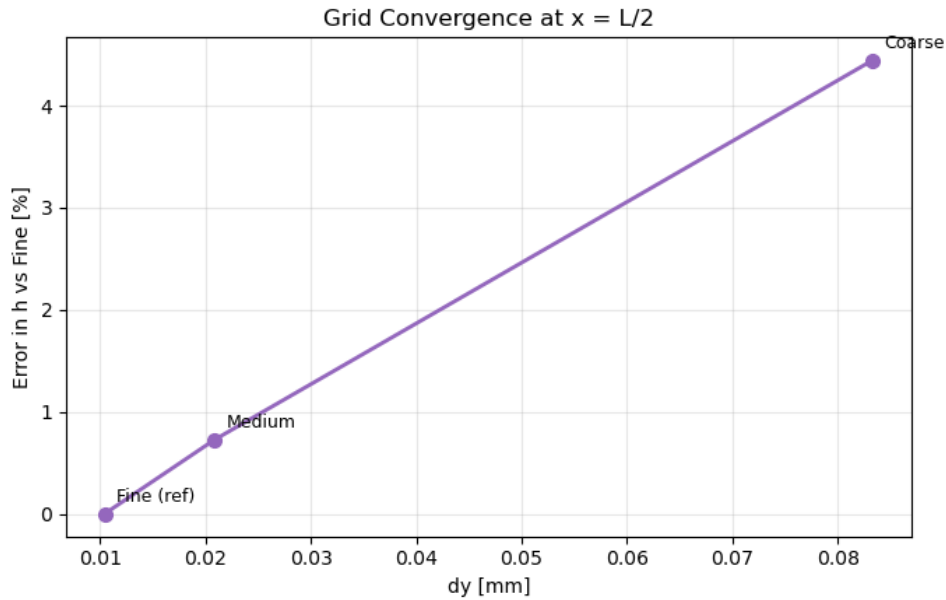


Figure 4.3. Grid convergence study: profiles at $x = L/2$ for the three mesh levels (coarse, medium, fine) converging toward the analytical solution.

4.4 Biot Analysis for Chip

With the convective heat transfer coefficient h now known from the boundary-layer analysis, the Biot number for the chip can be computed. Using the chip's half-thickness as the characteristic length ($L_c = 5 \text{ mm}$) and the silicon thermal conductivity ($k_{\text{Si}} = 148 \text{ W/(m K)}$):

$$\text{Bi} = h L_c / k_{\text{Si}} = 1.10 \times 10^{-4} \ll 0.1$$

This extremely small Biot number (four orders of magnitude below the threshold) confirms that the internal temperature gradients within the silicon die are negligible compared to the external convective resistance. The chip can therefore be treated as a lumped body in room-scale simulations, consistent with the finding from Project 2, Section 3.3.

4.4.1 Darcy-Forchheimer Setup for Porous Medium Rack Modeling

The boundary-layer results also inform the parameterization of server racks as porous media in room-scale CFD. The Darcy-Forchheimer model treats the rack as a distributed resistance to airflow, with a pressure drop that depends on both a viscous (Darcy) term and an inertial (Forchheimer) term:

$$-dP/dx = (\mu / K) v + C_F \rho v^2$$

where K is the permeability, C_F is the Forchheimer coefficient, and v is the superficial velocity through the rack. The chip-level convection coefficient from the

Blasius-Pohlhausen analysis feeds into the volumetric heat transfer coefficient used in the porous-medium energy equation, closing the link between the chip-scale physics (Projects 2 - 3) and the room-scale simulation (Project 5).

CHAPTER 5

COMSOL Multiphysics Simulation

This chapter bridges the gap between the hand-coded Python solver of Project 3 and commercial-grade simulation tools by reproducing the flat-plate boundary-layer problem in COMSOL Multiphysics. The goal is twofold: to validate the Python results against an independent solver, and to assess the effect of mesh refinement on solution accuracy and computational cost.

5.1 Simulation Setup

The computational domain is a two-dimensional rectangle of 60 mm x 30 mm, consisting of a 10 mm adiabatic pre-chip section (to allow the velocity boundary layer to develop before the heated region) and a 50 mm chip surface with constant wall temperature $T_{\text{wall}} = 350$ K. The free-stream conditions are $U_{\text{inf}} = 1$ m/s and $T_{\text{inf}} = 300$ K, giving a chip Reynolds number of $Re_L = 3333$.

5.1.1 Boundary Conditions

Boundary	Velocity BC	Thermal BC
Inlet (left)	$U = 1$ m/s, $V = 0$	$T = 300$ K
Outlet (right)	$dU/dx = 0$, $p = 0$	$dT/dx = 0$
Top (far-field)	Slip (symmetry)	$T = 300$ K
Pre-chip wall (0-10 mm)	No-slip ($U = V = 0$)	Adiabatic ($dT/dn = 0$)
Chip wall (10-60 mm)	No-slip ($U = V = 0$)	$T = 350$ K

5.1.2 Mesh Details

Two meshes are investigated to quantify the sensitivity of the solution to spatial resolution:

Mesh	Elements	Min. Element Size	Boundary Layers	Wall Clock Time
Uniform	41,096	0.73 mm	No	129 s
Refined	250,051	0.02 mm	15 layers	150 s

The refined mesh uses 15 structured boundary-layer elements near the wall, with a first cell height of 0.02 mm and a growth ratio of 1.2. This ensures that the steep velocity and temperature gradients near the wall are well resolved. The physics modules used are Laminar Flow (spf) and Heat Transfer in Fluids (ht), coupled via the Non-Isothermal Flow

multiphysics interface.

5.2 Velocity and Temperature Fields

Figure 5.1 compares the COMSOL-computed velocity profiles with the Blasius analytical solution at several streamwise locations along the chip. The profiles are plotted in Blasius similarity coordinates (η vs. f') to test whether self-similarity is achieved.

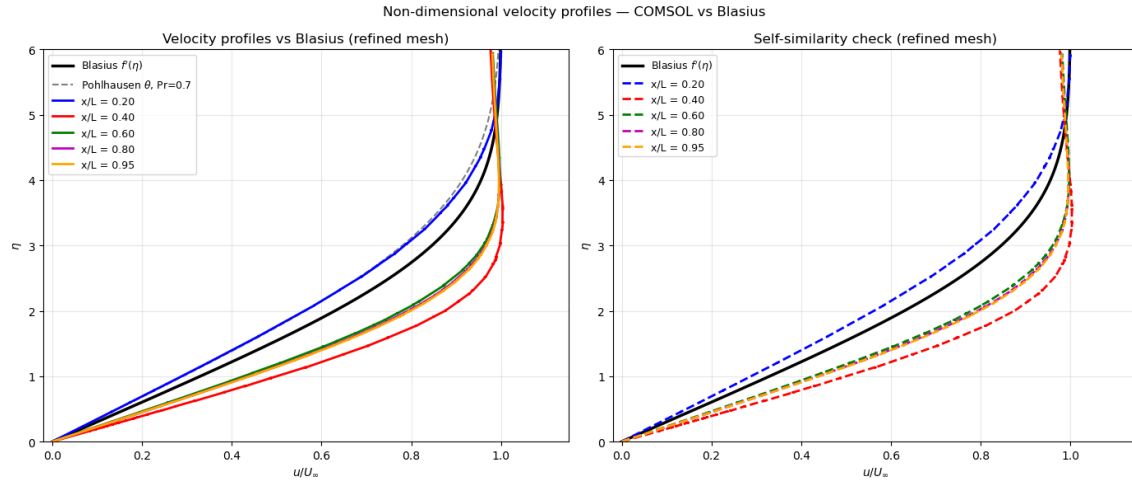


Figure 5.1. COMSOL velocity profiles (both uniform and refined meshes) compared to the Blasius analytical solution in similarity coordinates. Both meshes show excellent collapse onto the analytical curve, confirming self-similarity.

Both meshes produce velocity profiles that collapse well onto the Blasius curve, confirming that self-similarity is achieved over the chip surface. The refined mesh shows marginally better agreement very close to the wall ($\eta < 1$), where the velocity gradient is steepest, but the difference is small — indicating that even the uniform mesh is adequate for capturing the overall boundary-layer structure.

The temperature field (not shown separately) exhibits the expected Pohlhausen profile, with the thermal boundary layer slightly thinner than the velocity boundary layer due to $Pr = 0.7 < 1$. The temperature transitions smoothly from $T_{\text{wall}} = 350$ K at the chip surface to $T_{\text{inf}} = 300$ K in the free stream.

5.3 Local Heat Flux and Nusselt Number

Figure 5.2 shows the local wall heat flux and the corresponding Nusselt number as functions of the streamwise coordinate x along the chip. These are the quantities of greatest practical interest, as they determine the rate of heat removal from the chip surface.

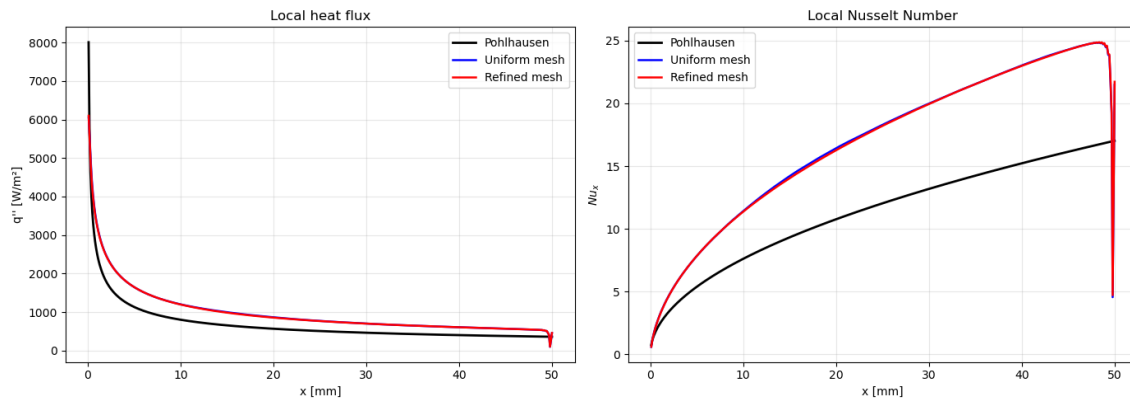


Figure 5.2. Local heat flux (left axis) and Nusselt number (right axis) along the chip surface, comparing COMSOL results (both meshes) with the Pohlhausen analytical solution.

Several observations emerge from Figure 5.2. First, both meshes agree well with the Pohlhausen analytical solution across the entire chip length, validating the COMSOL setup. Second, the COMSOL Nusselt numbers are systematically slightly above the analytical values. This is a well-known finite-domain effect: the COMSOL domain has a finite height (30 mm), which constrains the boundary layer and slightly accelerates the free-stream flow above it, enhancing heat transfer. In an infinite domain (as assumed by the analytical theory), this confinement effect vanishes.

Third, the heat flux is highest at the leading edge of the chip ($x = 0$ mm) and decreases monotonically downstream, following the characteristic $x^{-1/2}$ scaling of laminar boundary layers. This is because the thermal boundary layer thickens with distance from the leading edge, increasing the thermal resistance between the wall and the free stream. The practical implication for chip design is that the leading edge of a processor die is the most effective location for heat removal.

5.4 Mesh Comparison and Computational Cost

The two meshes differ by a factor of 6 in element count (41,096 vs. 250,051) but only 16 percent in wall-clock time (129 s vs. 150 s). This favorable scaling is a consequence of COMSOL's direct solver (MUMPS), which is relatively insensitive to element count for 2D problems in this size range.

The practical conclusion is that for this particular problem — a simple 2D laminar boundary layer — even the coarser uniform mesh provides engineering-grade accuracy (errors below 5 percent in Nu_x). However, for more complex geometries (e.g., turbulent flows, recirculation zones, or conjugate heat transfer with solid conduction), boundary-layer mesh refinement becomes essential. The 15-layer refined mesh used here represents best practice for wall-bounded thermal simulations and would be the recommended starting point for any production-quality COMSOL study.

CHAPTER 6

Data Hall Simulation with Reality DC

The final project returns to the room scale, applying all of the earlier insights in a realistic simulation of a data hall using the Reality DC platform. This chapter progresses from a baseline open-aisle layout, through cold-aisle containment, to adaptive fan-speed control — demonstrating the dramatic improvements in thermal performance and energy efficiency achievable through thoughtful design.

6.1 Baseline Model Setup

The data hall is an 8.25 m x 8.25 m room containing 12 server cabinets arranged in two rows of six, forming a hot-aisle/cold-aisle configuration. Each cabinet dissipates 5 kW of heat, for a total IT load of 60 kW. Cooling is provided by Air Cooling Units (ACUs) that supply chilled air through an underfloor plenum and perforated floor tiles in the cold aisle.

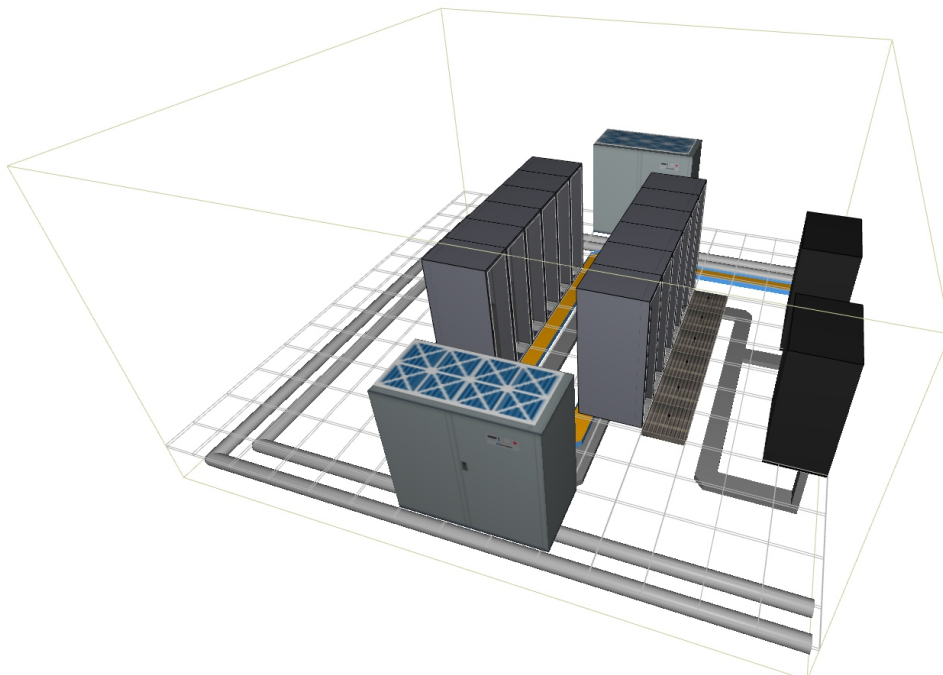


Figure 6.1. Isometric view of the baseline data hall in Reality DC Design Pro: 12 server cabinets in two rows, two ACUs (front and back), two PDUs (left), underfloor cooling pipes (grey) and perforated cold-aisle tiles (orange).

6.1.1 Component Inventory

Component	Quantity	Key Parameter
Server Cabinets	12 (2 x 6)	5 kW each, 60 kW total
ACUs	N+1 redundancy	$T_{\text{supply}} = 18$ degrees C setpoint
PDUs	N+1 redundancy	Electrical distribution
Perforated Tiles	12	25% open area, cold aisle
Underfloor Plenum	Full room	300 mm depth
Underfloor Pipes	CHW supply/return	Connected to ACUs

The ACUs operate at 100 percent fan speed in the baseline configuration, with a cubic power-law relationship between fan speed and electrical power consumption: $W_{\text{fan}} = W_{\text{rated}} \times (N/N_{\text{max}})^3$. The supply air temperature setpoint is 18 degrees C.

6.2 Baseline Results

The baseline simulation reveals significant hot-air recirculation in the open-aisle configuration. Hot exhaust air from the server cabinets (at approximately 35 degrees C in the hot aisle) spills over the tops of the cabinets and mixes with the cold supply air in the cold aisle, raising the cabinet inlet temperatures above the intended 18 degrees C supply temperature.

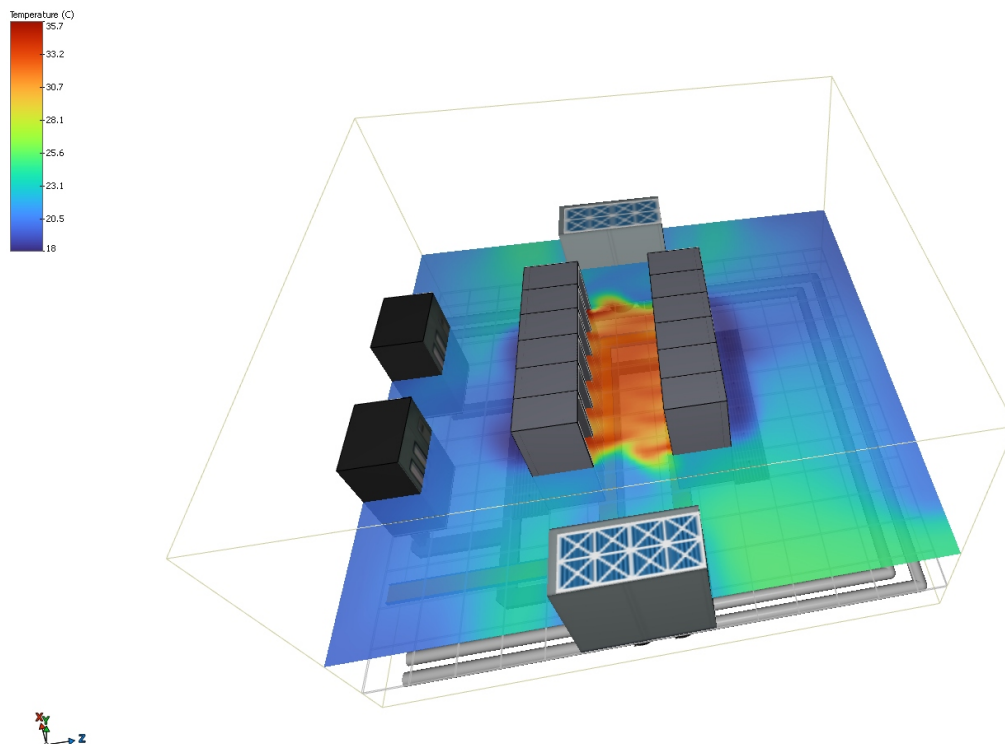


Figure 6.2. Baseline temperature field at 1 m above the floor (Reality DC). Peak temperature reaches 35.7 °C directly above the hot aisle (red zone); cold supply air (blue) is visible near the ACUs and perforated tiles.

Key baseline results include:

- Peak temperature at 1 m plane: approximately 35.7 °C (hot aisle)
- Maximum cabinet inlet temperature: 29.1 °C
- Minimum cabinet inlet temperature: approximately 20 °C (cabinets nearest ACUs)
- ASHRAE 2021 A1 compliance: several cabinets in amber/red zone (inlet > 27 °C)

The 11 °C spread in cabinet inlet temperatures (from 20 to 29.1 °C) indicates poor airflow management. Cabinets far from the ACUs receive pre-heated air, creating hot spots that can reduce server reliability and trigger throttling. This baseline motivates the implementation of containment strategies.

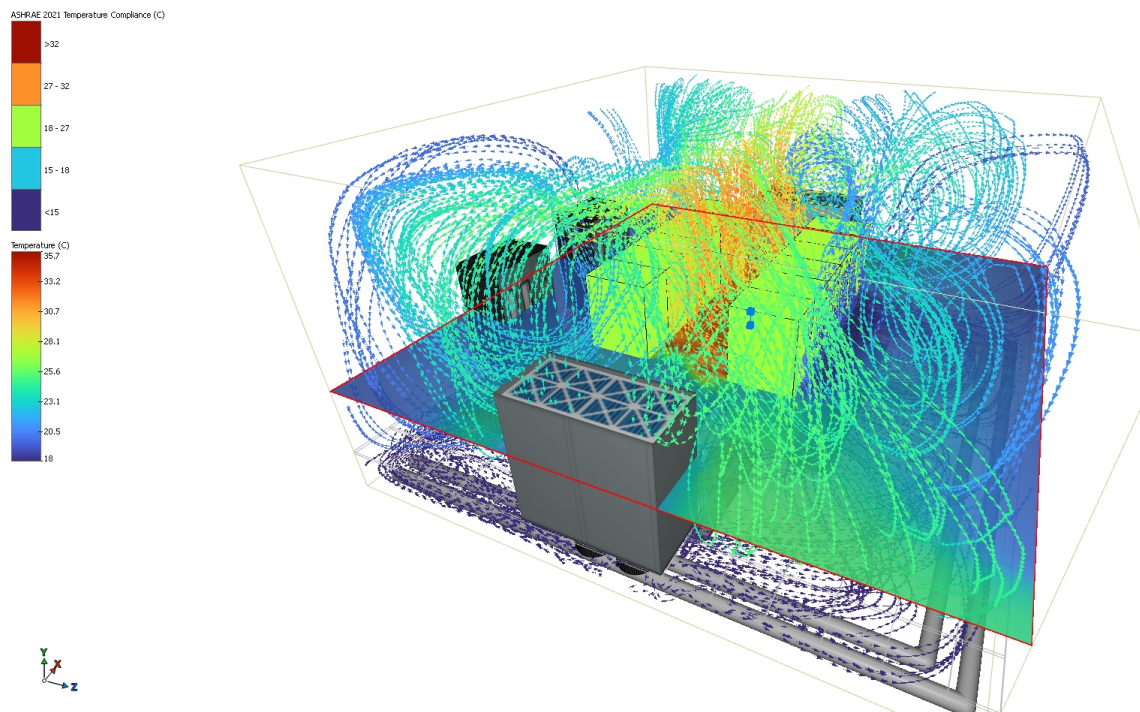


Figure 6.3. Baseline ASHRAE 2021 temperature compliance and room streamlines. Streamlines colored by temperature (18–35.7 °C) reveal strong hot-air recirculation loops over the cabinet rows; several cabinets exceed the 27 °C recommended limit.

6.3 Cold Aisle Containment

Cold-aisle containment (CAC) encloses the cold aisle with physical barriers (end-of-row doors and a ceiling panel) that prevent hot exhaust air from mixing with the supply air. In the Reality DC model, containment is implemented by adding aisle enclosures and removing the

cabinet legs (which would otherwise allow air to bypass the containment under the cabinets).

The results are dramatic. With containment in place, the maximum cabinet inlet temperature drops from 29.1 degrees C to 18.0 degrees C — exactly equal to the ACU supply setpoint. This means that every cabinet receives air at the designed supply temperature, with zero hot-air recirculation.

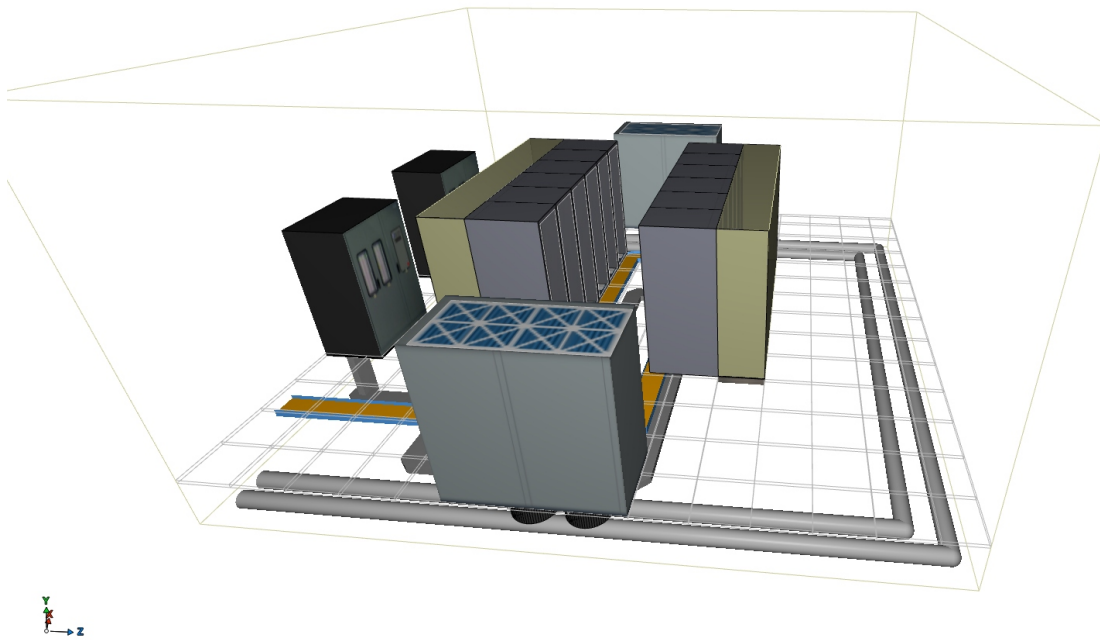


Figure 6.4. Isometric view with cold-aisle containment in place (tan enclosure panels). Cabinet legs have been removed to eliminate air bypass under the containment.

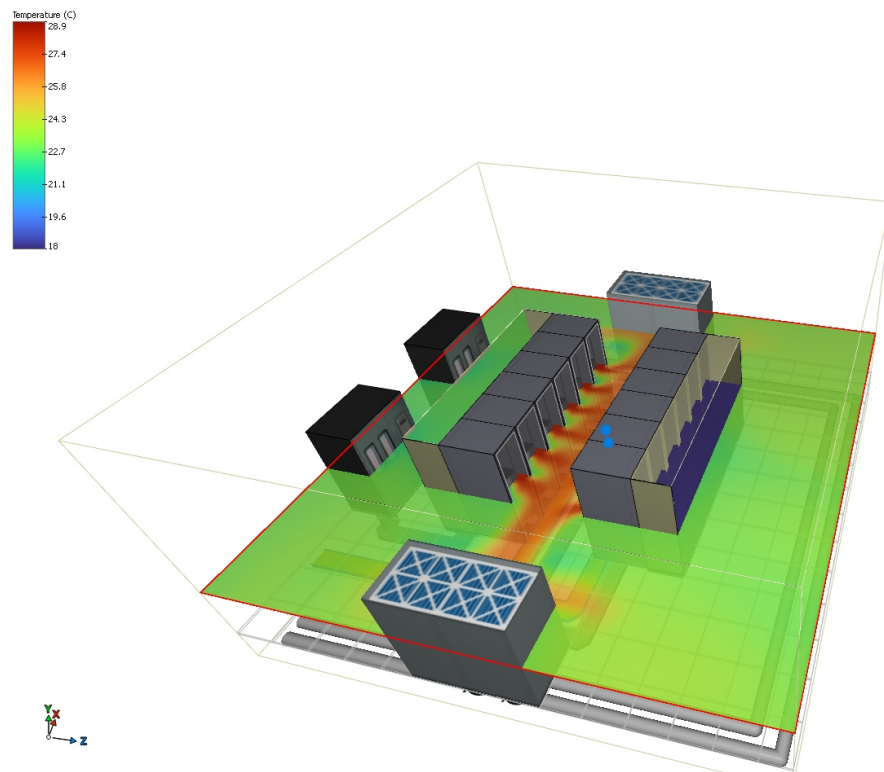


Figure 6.5. Temperature field at 1 m with cold-aisle containment (same scale as baseline). Peak temperature drops from 35.7 °C to 28.9 °C; the room outside the aisle becomes uniformly warm (~21–22 °C, green).

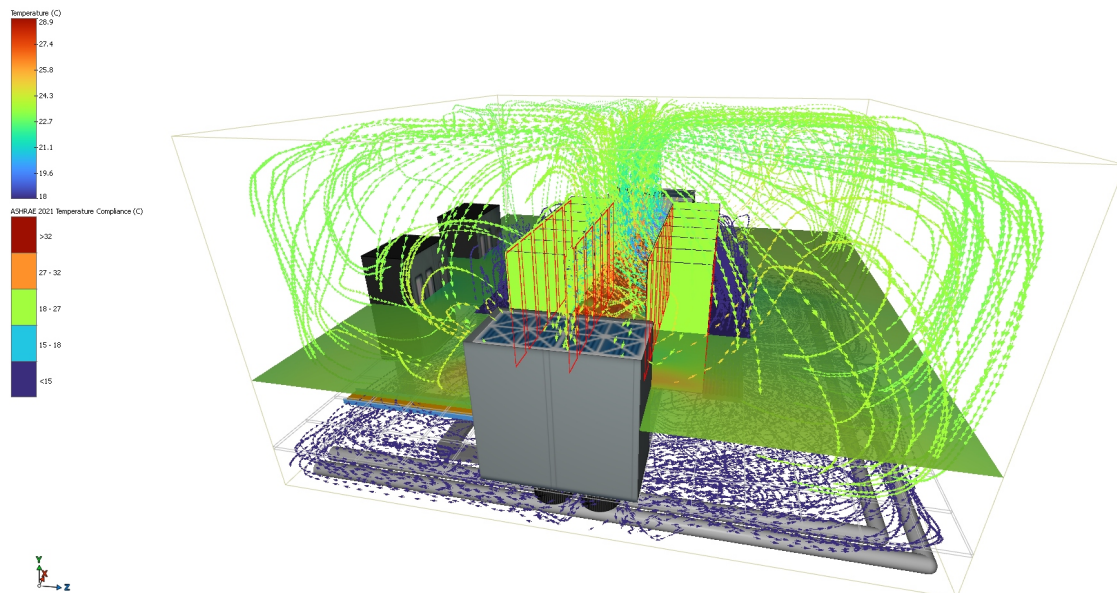


Figure 6.6. ASHRAE 2021 compliance with containment: all streamlines are green/blue (18–27 °C), confirming zero hot-air recirculation into the cold aisle. Every cabinet inlet is at exactly 18 °C (supply setpoint).

6.3.1 Performance Metrics

Metric	Value	Interpretation
Supply Heat Index (SHI)	0.0008	Negligible hot-air recirculation
Return Heat Index (RHI)	0.999	Nearly all heat removed by ACUs
Rack Temperature Index (RTI)	55.5	Good thermal isolation
Room Efficiency	84.3%	High fraction of cooling reaching racks
PUE	1.186	Excellent energy efficiency
Max Cabinet Inlet	18.0 degrees C	= supply setpoint (ideal)
ASHRAE 2021 Compliance	100%	All cabinets in green zone

The Supply Heat Index (SHI) of 0.0008 quantifies the fraction of heat that recirculates from the hot aisle to the cold aisle — essentially zero. The Return Heat Index (RHI) of 0.999 confirms that virtually all of the IT heat is captured and removed by the ACUs. The PUE of 1.186 means that for every watt consumed by the IT equipment, only 0.186 watts are consumed by the cooling infrastructure — an excellent result for an air-cooled facility.

One notable side effect of containment is that the room outside the contained aisle becomes warmer, because the hot exhaust air no longer disperses throughout the room. This is actually desirable from a thermodynamic perspective: higher return-air temperatures to the ACU coils improve the chiller's COP (linking back to the analysis in Project 1), and the contained hot aisle can be allowed to run hotter without affecting server inlet temperatures.

6.4 ACU Fan Speed Control

With containment in place, the next optimization step is to implement adaptive fan-speed control on the ACUs. Two scenarios are investigated.

6.4.1 Task 3.1 — 5 kW per Cabinet with Temperature Control

A temperature controller is implemented with a return-air setpoint of 30 degrees C. At the baseline load of 5 kW per cabinet (60 kW total), the controller finds that the current fan speed already satisfies the setpoint — no reduction in fan speed is possible without exceeding the return-air temperature limit. The operating point is unchanged.

6.4.2 Task 3.2 — 7.5 kW per Cabinet (90 kW Total)

The cabinet load is increased by 50 percent to 7.5 kW each (90 kW total) to test the system's capacity under higher-density computing loads. With cold-aisle containment and the ACUs running at full speed, the results are:

Parameter	5 kW/Cabinet	7.5 kW/Cabinet
Total IT Load	60 kW	90 kW
Max Cabinet Inlet	18.0 degrees C	21.0 degrees C
ASHRAE Safety Margin	9 K	11 K
ASHRAE 2021 Compliance	100%	100%
Fan Speed	100%	100%
Containment	Yes	Yes

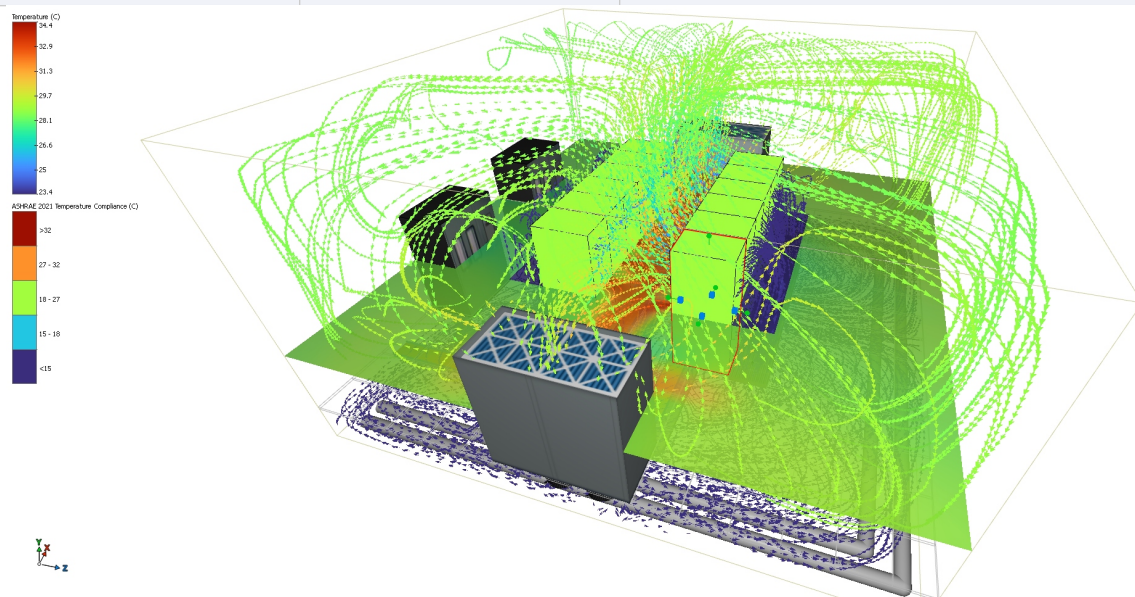


Figure 6.7. ASHRAE 2021 compliance at 5 kW/cabinet with temperature-controlled ACUs (30 °C return setpoint). All cabinets remain in the green compliance band; max inlet = 23.5 °C.

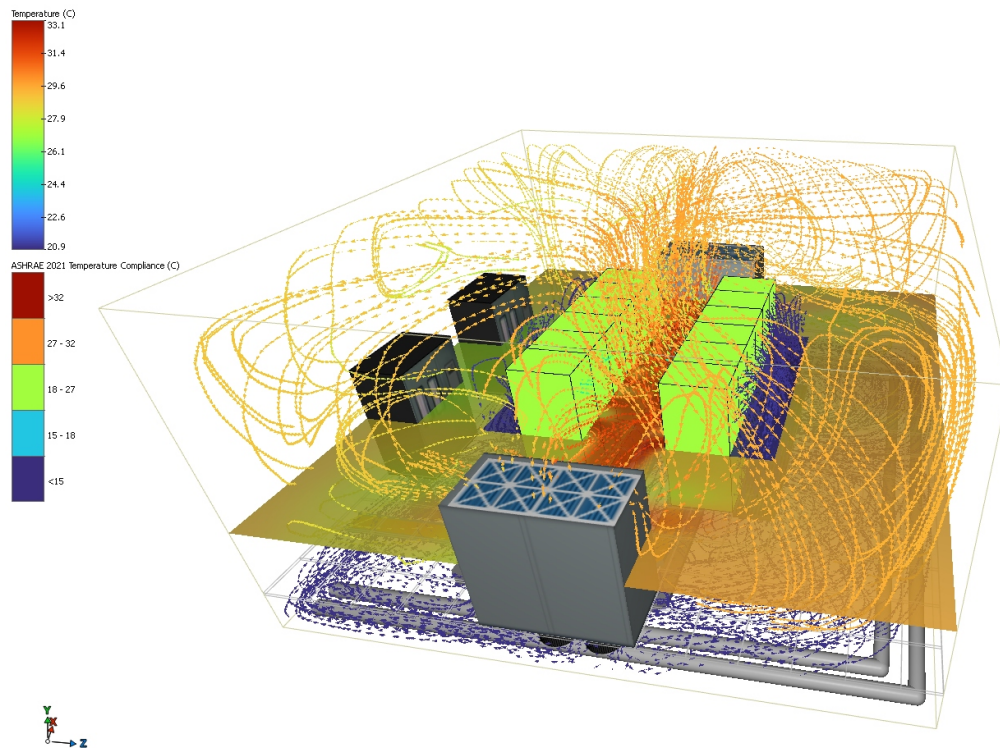


Figure 6.8. ASHRAE 2021 compliance at 7.5 kW/cabinet (90 kW total). Despite a 50% increase in IT load, all cabinets remain compliant; max inlet = 21 °C, safety margin = 11 K above the ASHRAE A1 limit.

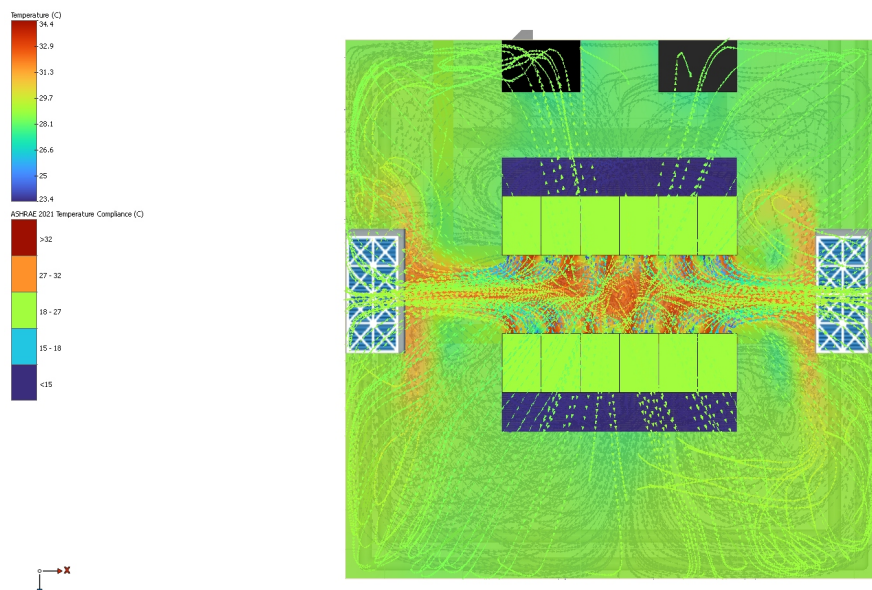


Figure 6.9. Top-view of the room at 7.5 kW with temperature controller: streamlines show well-organized airflow between ACUs and cabinet rows, with no significant recirculation despite the higher heat load.

Even with a 50 percent increase in heat load, the maximum cabinet inlet temperature rises only to 21 degrees C — still 11 K below the ASHRAE 2021 A1 recommended limit of 27 degrees C. This demonstrates that the combination of cold-aisle containment and adequately sized ACUs provides substantial headroom for load growth. The containment eliminates the recirculation that would otherwise cause hot spots, allowing the increased heat to be uniformly distributed across all cabinets.

In a real facility, this headroom could be exploited in two ways: (1) the ACU fan speed could be reduced (saving energy via the cubic power law) while still maintaining acceptable inlet temperatures, or (2) the cabinet density could be increased further to accommodate higher-power AI accelerators without infrastructure upgrades. This flexibility is one of the most important benefits of proper airflow management.

CHAPTER 7

Cross-Scale Integration and Design Implications

7.1 The Multi-Scale Pipeline

The five projects in this portfolio are not independent exercises — they form a coherent multi-scale pipeline in which the output of each project feeds the next. The connections are as follows:

- Project 1 to Project 5: The system COP curve from Project 1 contextualizes the PUE from Project 5. The PUE of 1.186 achieved with cold-aisle containment implies a cooling system COP of approximately $1/0.186 = 5.4$. Referring to the COP curve in Figure 2.1, this corresponds to a wet-bulb temperature of approximately 25 degrees C — meaning the Reality DC result is consistent with a moderately warm operating environment.
- Project 2 to Project 3: The conduction analysis in Project 2 reveals the internal temperature distribution within the chip package. The boundary-layer analysis in Project 3 provides the convective coefficient h needed to compute the Biot number, which in turn determines whether the lumped-body simplification used in Projects 2.3 and 5 is valid.
- Project 3 to Project 4: The Blasius-Pohlhausen analytical solution from Project 3 serves as the benchmark for the COMSOL simulation in Project 4. The excellent agreement validates both the analytical theory and the COMSOL setup.
- Project 3 to Project 5: The Darcy-Forchheimer porous-medium parameters derived from the boundary-layer analysis feed directly into the server-rack model in Reality DC, linking the chip-scale convection physics to the room-scale airflow simulation.
- Project 4 to Project 5: The mesh-sensitivity analysis in COMSOL (Chapter 5) informs the mesh requirements for the Reality DC simulation. The finding that boundary-layer refinement improves near-wall gradients but has minimal impact on bulk flow quantities justifies the use of moderate-resolution meshes in room-scale CFD.

7.2 The Multi-Scale Picture

The spatial scales addressed in this portfolio span six orders of magnitude:

Scale	Characteristic Length	Project	Key Physics
Transistor junction	10 - 100 nm	—	Joule heating, quantum effects

Scale	Characteristic Length	Project	Key Physics
Die / TIM	0.5 - 10 mm	P2	Conduction, contact resistance
Boundary layer	0.1 - 5 mm	P3, P4	Laminar convection, Blasius-Pohlhausen
Server / rack	0.5 - 2 m	P5	Porous-medium flow, Darcy-Forchheimer
Data hall	8.25 m	P5	Room CFD, recirculation, containment
Cooling plant	10 - 100 m	P1	Psychrometrics, COP, economizer
Atmosphere	km-scale	P1	Wet-bulb temperature, climate

At the smallest scales, heat is generated by billions of transistor switching events and must be conducted through the die and TIM layers — a process governed by Fourier's law and characterized by the thermal conductivities measured in Project 2. At the chip surface, the heat is transferred to the airflow via a convective boundary layer whose properties are described by the Blasius-Pohlhausen theory (Projects 3 - 4). Within the server rack, the airflow encounters a complex geometry of circuit boards, heat sinks, and cable bundles, which is most efficiently modeled as a porous medium. At the room scale (Project 5), the interactions between dozens of racks, ACUs, and floor tiles create recirculation patterns that determine whether servers receive adequately cool air. And at the plant scale (Project 1), the entire cooling chain — from chilled-water production to atmospheric heat rejection — determines the energy cost of every joule removed from every chip.

7.3 Design Recommendations

Based on the comprehensive analysis across all five projects, the following design recommendations emerge for energy-efficient data center cooling:

- **Maximize free-cooling hours:** The COP curve from Project 1 shows that free cooling (Mode 1) achieves COP values 5 to 10 times higher than mechanical cooling (Mode 3). Site selection in cool, dry climates is the single most impactful decision for cooling efficiency.
- **Implement cold-aisle containment:** Project 5 demonstrates that containment eliminates hot-air recirculation, reducing maximum cabinet inlet temperatures by 11 degrees C and achieving a PUE of 1.186. This is a relatively low-cost retrofit with massive benefits.
- **Use adaptive fan control:** The cubic power law for fan energy means that reducing fan speed by 20 percent cuts fan power by nearly 50 percent. With containment in place, significant fan-speed reductions are possible without exceeding ASHRAE limits.
- **Invest in chip-level thermal management:** The 2D conduction analysis in Project 2 shows that TIM resistance is a critical bottleneck. Advanced TIMs (metallic, diamond-loaded) can reduce the die-to-enclosure temperature drop, enabling higher power densities.

- Validate with multi-scale analysis: No single model captures all the relevant physics. System-level COP models (Project 1) miss room-scale hot spots; room-scale CFD (Project 5) misses chip-level thermal bottlenecks. A multi-scale approach, like this portfolio, is essential for a complete understanding.

7.4 Summary of Key Numerical Results

The following table compiles the most important quantitative results from all five projects, providing a single reference for the key findings of this portfolio.

Project	Quantity	Value	Significance
P1	COP range	3 - 30+	10x variation with climate
P1	BP1 / BP2	7 / 12 degrees C	Mode transition wet-bulb T
P2	L_2 error (t=10s)	4.00%	1D solver convergence
P2	Grid (2D)	181 x 65	Chip temperature resolution
P2	Bi (steel rod)	< 0.1	Lumped body valid
P3	$f''(0)$	0.332057	Blasius wall shear
P3	$\theta'(0)$	0.2927	Pohlhausen wall T gradient
P3	Grid convergence	2nd order	CFD verification
P4	Mesh elements	41k / 250k	Uniform / refined
P4	Wall time	129 / 150 s	COMSOL solve time
P4	Bi (chip)	1.10×10^{-4}	Lumped body confirmed
P5	Baseline max inlet	29.1 degrees C	Hot-air recirculation
P5	CAC max inlet	18.0 degrees C	= supply setpoint
P5	SHI	0.0008	Zero recirculation
P5	RHI	0.999	Complete heat capture
P5	PUE	1.186	Excellent efficiency
P5	7.5 kW margin	11 K	ASHRAE headroom

Taken together, these results demonstrate that rigorous multi-scale thermal analysis — from the chip junction to the cooling tower — is not merely an academic exercise but a practical necessity for the design and operation of modern data centers. As IT power

densities continue to rise with the proliferation of AI accelerators, and as climate change raises ambient temperatures globally, the coupling between these scales will only grow stronger. The tools and methods demonstrated in this portfolio — analytical solutions, finite-difference solvers, commercial FEM software, and room-scale CFD platforms — provide a comprehensive toolkit for addressing these challenges.